

Claudio César Claros Olivares

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Education

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| Ph.D. University of Delaware , Electrical and Computer Engineering
<ul style="list-style-type: none"> Advisor: Austin J. Brockmeier, Computational Neural and Information Engineering Lab Research: reliable machine learning under distribution shift (out-of-distribution detection, calibration) and machine learning for brain imaging and sports medicine Bendett Fellowship (2023); Best Poster Award, ECE Research Day (2024) | Newark, DE, USA
Jan 2022 – present |
| M.Sc. University of Delaware , Electrical and Computer Engineering
<ul style="list-style-type: none"> Concentration: Signal Processing, Communications, and Controls Thesis: <i>Synergistic Human-Machine Prediction: Active Error Analysis and Mitigation with Gaussian Process Regression</i> (advisor: Austin J. Brockmeier) GPA: 3.93/4.00 | Newark, DE, USA
Aug 2018 – Aug 2020 |
| Certificate Centro Psicopedagógico y de Investigación en Educación Superior (CEPIES) , University Teaching | La Paz, Bolivia
Aug 2015 – Nov 2015 |
| B.S. Universidad Mayor de San Andrés (UMSA) , Electronics Engineering
<ul style="list-style-type: none"> Major: Control Systems Graduation project: <i>Prototipo para generación de consignas de control a partir de los ritmos cerebrales alpha y mu. Caso de estudio: Factibilidad de uso de la plataforma Emotiv EPOC</i> (advisor: Javier Sanabria García) Class valedictorian | La Paz, Bolivia
Feb 2007 – Dec 2014 |

Experience

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| Computational Neural and Information Engineering Lab, University of Delaware , Graduate Research Assistant
<ul style="list-style-type: none"> Benchmarked out-of-distribution detection methods across CNN and ViT representations and training paradigms, and derived worst-case calibration error bounds under distribution shift. Developed 3D convolutional networks and interpretation methods for brain age prediction from magnetic resonance elastography, in collaboration with Curtis L. Johnson's lab. Built an interpretable post-concussion musculoskeletal injury risk model from longitudinal clinical data, in collaboration with Thomas A. Buckley's lab. Ongoing: early prediction of cognitive decline on ADNI for the Delaware Longitudinal Study for Alzheimer's Prevention (Delaware Center for Cognitive Aging Research), and interpretable EEG biomarkers for epilepsy from routine scalp EEG with Nemours Children's Health. | Newark, DE, USA
Jan 2022 – present
4 years 9 months |
| Computational Neural and Information Engineering Lab, University of Delaware , Research Assistant
<ul style="list-style-type: none"> Developed a synergistic system for human-machine prediction through active error analysis of a model's systematic errors using Gaussian process regression. | Newark, DE, USA
July 2019 – Aug 2020
1 year 2 months |

Institute of Applied Electronics, UMSA , Researcher Designed and implemented embedded software for signal acquisition and control of a mobile robot prototype for multi-agent research.	La Paz, Bolivia June 2017 – Dec 2017 7 months
Institute of Applied Electronics, UMSA , Researcher Developed a platform of mobile sensor nodes for controlled environments, covering the design of the robotic units and the data acquisition and signal processing tools.	La Paz, Bolivia Apr 2016 – Dec 2016 9 months
Agencia para el Desarrollo de la Sociedad de la Información en Bolivia (ADSIB) , Research Intern Designed and implemented an FPGA interface for Ethernet communication dedicated to packet acquisition and routing for the FPGA-based router project.	La Paz, Bolivia Sept 2015 – Dec 2015 4 months
Digital Signal Processing Lab, Electronics Engineering Department, UMSA , Laboratory Assistant Responsible for the maintenance and operation of the digital signal processing laboratory.	La Paz, Bolivia July 2013 – Dec 2013 6 months
Institute of Applied Electronics, UMSA , Research Assistant Developed a software application for EEG acquisition and spectral analysis via the fast Fourier transform.	La Paz, Bolivia Mar 2012 – Aug 2012 6 months

Publications

Bounding Worst-Case Calibration Error in OOD Detection Under Distribution Shift Claudio C. Claros-Olivares, Austin J. Brockmeier openreview.net/forum?id=aZnAoyzssl (AISTATS 2026 Workshop on Calibration for Modern AI)	May 2026
A Systematic Analysis of Out-of-Distribution Detection Under Representation and Training Paradigm Shifts Claudio César Claros Olivares, Austin J. Brockmeier arxiv.org/abs/2511.11934 (arXiv preprint arXiv:2511.11934 (under review))	Nov 2025
A Machine Learning Model for Post-Concussion Musculoskeletal Injury Risk in Collegiate Athletes Claudio C. Claros, Melissa N. Anderson, Wei Qian, Austin J. Brockmeier, Thomas A. Buckley 10.1007/s40279-025-02196-4 (Sports Medicine, 55(8), 1971-1982)	Jan 2025
MRI-based whole-brain elastography and volumetric measurements to predict brain age Claudio C. Claros-Olivares, Rebecca G. Clements, Grace McIlvain, Curtis L. Johnson, Austin J. Brockmeier 10.1093/biomethods/bpae086 (Biology Methods and Protocols, 10(1), bpae086)	Jan 2024
Integrative data analysis to identify persistent post-concussion deficits and subsequent musculoskeletal injury risk: project structure and methods Melissa Anderson, Claudio Cesar Claros, Wei Qian, Austin Brockmeier, Thomas A. Buckley 10.1136/bmjsem-2023-001859 (BMJ Open Sport & Exercise Medicine, 10(1), e001859)	Jan 2024
Mechanical Property Based Brain Age Prediction using Convolutional Neural Networks Rebecca G. Clements, Claudio C. Claros-Olivares, Grace McIlvain, Austin J. Brockmeier, Curtis L. Johnson 10.1101/2023.02.12.528186 (bioRxiv)	Jan 2023

A Machine Learning Approach for Modeling and Inverse Design of Polymer Blends Soham Jariwala, Claudio César Claros Olivares, Zijie Wu bpb-us-w2.wpmucdn.com/sites.udel.edu/dist/d/12003/files/2022/05/Final_Dupont.postfeedback.pdf (Project report, CHEG/CIS 867, University of Delaware (industry mentor: DuPont))	May 2022
Identifying the Instances Associated with Distribution Shifts using the Max-Sliced Bures Divergence Austin J. Brockmeier, Claudio César Claros Olivares, Matthew S. Emigh, Luis G. Sanchez Giraldo neurips.cc/virtual/2021/35481 (NeurIPS 2021 Workshop on Distribution Shifts: Connecting Methods and Applications)	Dec 2021
Max-sliced Bures Distance for Interpreting Discrepancies Austin J. Brockmeier, Claudio Cesar Claros, Carlos H. Mendoza-Cardenas, Yüksel Karahan, Matthew S. Emigh, Luis Gonzalo Sanchez Giraldo openreview.net/forum?id=D2Fp_qheYu (Preprint, OpenReview)	Jan 2021
Synergistic human-machine prediction: Active error analysis and mitigation with Gaussian process regression Claudio César Claros Olivares (M.Sc. thesis, University of Delaware)	Jan 2020
Prototipo para generación de consignas de control a partir de los ritmos sensomotores. Caso de estudio: Factibilidad de empleo de la plataforma Emotiv Epoc Claudio César Claros Olivares, Javier Sanabria García (IEEE Bolivian Engineering and Technology Congress (BETCON), La Paz, Bolivia)	Jan 2015

Presentations

- **May 2026:** Poster, *Bounding Worst-Case Calibration Error in OOD Detection Under Distribution Shift*, AISTATS 2026 Workshop on Calibration for Modern AI, Tangier, Morocco.
- **May 2024:** Poster, ECE Research Day, University of Delaware. Best Poster Award, Signal Processing, Communication, and Control area.
- **Jul. 2024:** Co-author, *Predictors of Post-Concussion Musculoskeletal Injury Severity: An Integrated Data Analysis Approach*, UK Collaborating Centre on Injury and Illness Prevention in Sport (IOC Research Centre), Edinburgh, UK.
- **Jul. 2022:** Co-author, *Mechanical property-based brain age prediction using artificial neural networks*, World Congress of Biomechanics, Taipei, Taiwan.
- **Dec. 2021:** Poster, *Identifying the Instances Associated with Distribution Shifts using the Max-Sliced Bures Divergence*, NeurIPS 2021 Workshop on Distribution Shifts.
- **Feb. 2020:** Poster, *Active error analysis and mitigation for synergistic machine learning infrastructure*, DARWIN Computing Symposium, Data Science Institute, University of Delaware.
- **Nov. 2019:** Invited talk, *Learning an error-aware model for synergistic human-machine prediction*, SIG-AI Seminar, Department of Computer and Information Sciences, University of Delaware.
- **Apr. 2016:** Invited talk, *Digital signal processing algorithms and machine learning*, IEEE Seminar on Artificial Intelligence, IEEE Student Branch, Universidad Católica Boliviana, La Paz.
- **May 2015:** Invited talk on the *Sensor your Swarm* project, First International Scientific Congress on Engineering, La Paz.
- **Aug. 2014:** Invited talk, *Research on brain signal processing*, La Paz Ciudad Digital 2.0, representing the Institute of Applied Electronics, UMSA.

Honors & Awards

- **2024:** Best Poster Award (Signal Processing, Communication, and Control area), ECE Research Day, University of Delaware.
- **2023:** Bendett Fellowship, Department of Electrical and Computer Engineering, University of Delaware.
- **2017:** Bolivian Government Scholarship for graduate studies abroad (30 of 100 scholarships awarded nationwide that year).
- **2015:** Honorable Mention, *Most Challenging Team*, Minesweepers: Towards a Landmine-Free World 2015, Antofagasta, Chile (Bolivian Team).
- **2015:** Senatorial Declaration, Chamber of Senators of the Plurinational Legislative Assembly of Bolivia, for outstanding participation in the NASA Space Apps Challenge 2015.
- **2015:** Resolution of the Honorable University Council, UMSA, for outstanding participation in the NASA Space Apps Challenge 2015.
- **2015:** Worldwide finalist, NASA Space Apps Challenge 2015: nationwide winner for *Best Robotics Theme Project* and worldwide finalist for *Best Mission Concept Project* with the project *Sensor your Swarm*.
- **2015:** Maximum score on the undergraduate graduation project, Electronics Engineering Department, UMSA.
- **2010:** Remarkable academic performance, UMSA. Congratulatory letter awarded to the best students of the year.

Technical Skills

Programming languages: Python, C/C++, MATLAB, Octave, JavaScript, Julia

Machine learning: PyTorch, PyTorch Lightning, Hydra, TensorFlow, Keras, scikit-learn

Scientific computing: NumPy, SciPy, pandas, OpenCV, CUDA

Languages

Spanish

Native

English

Fluent (TOEFL 104/120, ECCE)

French

Intermediate (DELF B2)